

Section B (25 minutes test + 25 minutes GTA tutorial)

77.5%

Question 3Part (a)

For each of the materials in the table below,

- State the class of material and the type of atomic bonding. [20%]
- State which of the bonds are highly directional. [5%]
- State, giving your reasoning, which material you expect to have the highest
 - Melting point;
 - Stiffness;
 - Electrical conductivity (in solid state).

[15%]

Material	Relative bond energy
Ti	medium
MgO (similar to NaCl)	high
Polyethylene	low
Diamond	Very high

Part (b)

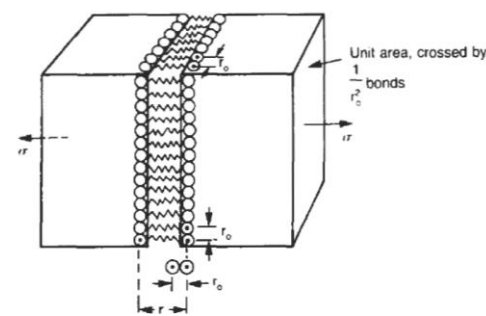
Copper has an atomic radius of 0.128 nm and an atomic weight of 63.5 g/mol. Calculate its theoretical density.

[15%]

Part (c)

If the atom bonds are stretched, one can imagine the bonds acting like springs between atoms in the elastic region as shown in the figure below. Stretch of the bonds gives the elastic strain. Using Hook's law, express and explain the physical origin of Young's modulus.

[20%]



(a) (i) Ti - metallic ~~band~~ (metal)

MgO - ionic ~~band~~ (salt) - 2.5%

Polythylene - ~~covalent band~~ ^{covalent bond in molecules and Van der Waal's among the molecules (polymer)} and Van der Waal's ~~band between layers~~ - 5%

Diamond - covalent ~~band~~ ^(ceramics) ~~(carbon)~~ - 2.5%

(ii) covalent ~~band~~

(iii) Diamond has the highest melting ~~point~~ and stiffness ~~as~~ its relative band energy is highest.

Ti has the highest electrical ~~conductivity~~ in solid state as it has metallic ~~band~~ with low relative band energy. ... - 2.5% ^(which have the sea of electrons)

(b) number of atoms = $(6 \times \frac{1}{2}) + (8 \times \frac{1}{8}) = 4$

$$\sqrt{2}a = 4r = 0.512 \text{ nm} = 5.12 \times 10^{-10} \text{ m}$$

$$\therefore \rho = \frac{4m}{V} = \frac{4 \times 6.022 \times 10^{23} \text{ mol}^{-1} \times 63.5 \text{ g/mol}}{a^3} \\ = \cancel{3.22 \times 10^5 \text{ kg/m}^3} \quad 8.9 \times 10^3 \text{ kg/m}^3$$

$$4 \times \frac{63.5 \text{ g/mol}}{6.022 \times 10^{23} \text{ mol}^{-1}} \\ a^3 \\ -10\%$$

(c) Hook's law : $E = \frac{6}{\epsilon}$

According to the diagram, $\phi = NS_0(r - r_0)$

$$= \frac{1}{r_0^2} S_0(r - r_0)$$

$\therefore$

$$= \frac{S_0}{r_0} \frac{r - r_0}{r_0}$$

$$= \epsilon \frac{S_0}{r_0}$$

$$E = \frac{\epsilon \frac{S_0}{r_0}}{\epsilon} = \frac{\epsilon_0}{r_0}$$

Part (d)

Sketch three-dimensional views of a cubic unit cell, showing

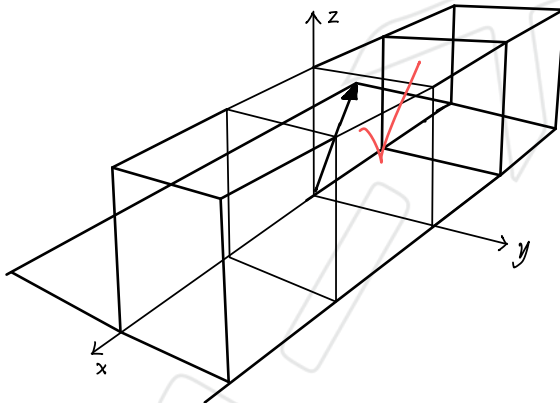
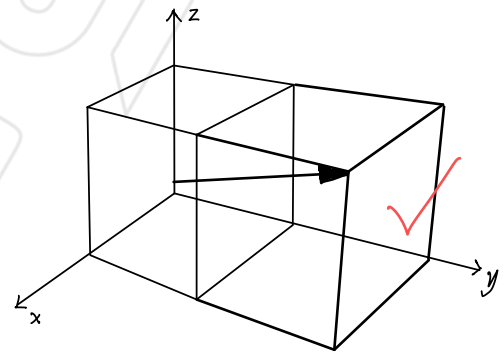
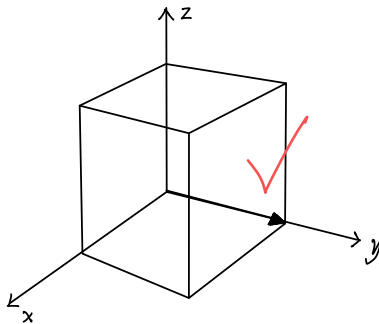
- i. a $[010]$ direction, a $[121]$ direction and a $[\bar{2}\bar{1}0]$ direction.

[15%]

- ii. What is the family of face diagonals?

[10%]

i.



ii. $\langle 0, 1, 1 \rangle$